College of Industrial Technology

King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Midterm Examination of Semester 2

Subject: 341152 Electric Circuits II

Date: 27 December 2013

Year: 2013

Section: 5-6

Time: 15.00-17.00

Name:______________________________ID:____________________Class:____________________

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Text books are **NOT** allowed.
4. A calculator is allowed.
5. Electronic communication devices are **NOT** allowed in the examination room.
6. The examination has 11 pages (including this page), 3 parts (55 questions) and a total score of 71 points.
7. Write all your solutions and answers on this examination and answer sheet.

Part 1 (25 points) True/False

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. In parallel network, total admittance is the sum of the individual admittance.
2. The time constant for an inductor circuit equals its resistance divided by its inductance.
3. An ideal inductor looks like a short circuit to dc current.
4. The steady-state level of the capacitor current can be found by substituting its open-circuit equivalent.
5. The heavier the current flow for a given voltage, the larger the value for admittance.
6. The voltage across a capacitor lags the current through it by 90°.
7. Inductive reactance increases inversely to frequency.
8. Power factor of resistive circuit is equal to one.
9. For a purely resistive element, the voltage and the current through the element are out of phase.
10. The current divider rule for dc circuits cannot be used for ac circuits.
11. For a series ac circuit with reactive elements, the voltages across a coil or capacitor are always in the same direction on a phasor diagram.
12. A capacitive network has a lagging power factor.
13. Total impedance of a system is the sum of the individual impedances.
14. Admittance is the reciprocal of impedance.
15. Capacitance is directly proportional to the distance between plates.
16. The magnitude of polar form is in RMS value.
17. When charging capacitor, the voltage of capacitor increases exponentially.
18. Resistor value is independent from frequency.

19. At very high frequency, capacitor acts as an open circuit.

20. In RC series circuit, electric current leads the voltage at power supply by 90 degrees.

21. When closing switch of RL DC circuit, electric current can flow in right at the beginning.

22. When charging inductor, the voltage of inductor increases exponentially.

23. The unit of inductive reactance is in Ohm.

24. The unit of reactive power is in VA.

25. The unit of power factor is in radian.

Part 2 (26 points) Multiple choices

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. Find the capacitance of a parallel-plate capacitor if a 35 V potential difference exists with 2,000 μC of charge.
   (a) 42 μF (b) 73 μF (c) 57 μF (d) 28 μF
2. Find the capacitance of a parallel-plate air dielectric capacitor if the area of each of its plates is 0.1 m^2 and the distance between the plates is 2.0 mm.
   (a) 1.44 nF (b) 0.62 nF (c) 0.44 nF (d) 0.14 nF

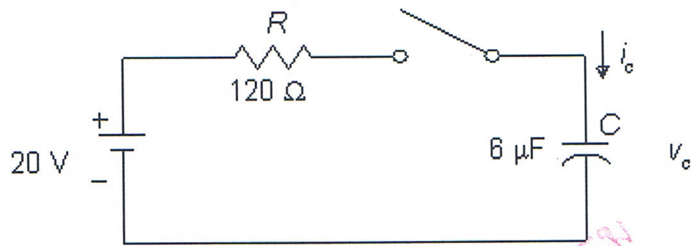


FIGURE 1

3. See Figure 1. What is the time constant in this circuit?
   (a) 720 μs (b) 820 μs (c) 780 μs (d) 450 μs
4. See Figure 1. After closing the switch, find the voltage v_C after 3 time constants.
   (a) 18.5 V (b) 19.3 V (c) 19 V (d) 17.8 V0 A

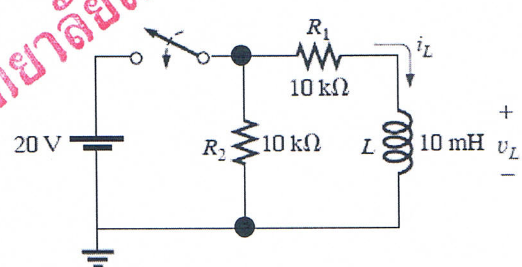


FIGURE 2

5. See Figure 2. After closing the switch, what will the voltage (v_L) across the inductor be after the circuit voltages and currents have reached steady-state values? Assume that the inductor is an ideal (lossless) device.
   (a) 0 V (b) 5 V (c) 10 V (d) 20 V

6. The currents and voltages are in steady state. Find V_1

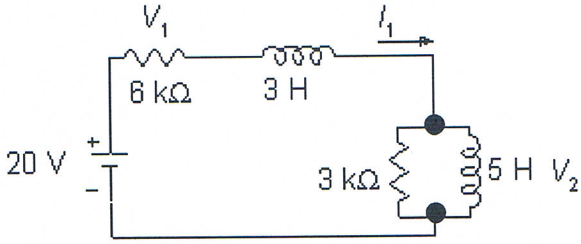


FIGURE 3

(a) 20 V (b) 18 V (c) 18.5 V (d) 20.5 V

7. See Figure 3. Find the current I_1 ?

(a) 2.33 mA (b) 3.67 mA (c) 3.33 mA (d) 2.67 mA

8. What is the inductive reactance at 0.5 Hz of a 1 H inductor?

(a) π Ω (b) 2π Ω (c) 1 Ω (d) 2 Ω

9. At what frequency does a $10/\pi$ μF capacitor have a reactance of 100 Ω?

(a) 50 Hz (b) 500 Hz (c) 500 kHz (d) 5 MHz

10. If the voltage v=50 sin(500t) is impressed across a 25 Ω resistor, which equation describes the resistor current?

(a) 2sin(500t + 90°) (b) 2cos(500t − 90°)

(c) 2 sin(500t) (d) 2sin(500t + 180°)

11. The voltage across a 100 mH coil is v=100 sin(50t). Which of these expressions describes the current?

(a) 2000 sin(50t − 90°) (b) 20 sin(50t + 90°)

(c) 20 sin 50t (d) 20 sin(50t − 90°)

12. Which one of the following polar values is equivalent to 30 − j40?

(a) 50 ∠36.9° (b) 50 ∠53.1° (c) 50 ∠-36.9° (d) 50 ∠-53.1°

13. Which one of the following values is equivalent to j^{10} ?

(a) 1 ∠0° (b) 1 ∠90° (c) 1 ∠180° (d) 1 ∠-90°

14. Which one of the following values is equivalent to −1 ∠180°?

(a) 1 ∠0° (b) 1 ∠90° (c) 1 ∠180° (d) 1 ∠-90°

15. Which one of the following rectangular values is equal to the polar form 20 ∠55°?

(a) 16.38 − j11.47 (b) 16.38 + j11.47

(c) 11.47 − j16.38 (d) 11.47 + j16.38

16. Which one of the following phasor domain expressions is equivalent to the time domain expression $50\sin(\omega t + 15°)$?

(a) $70.7\ \angle{-15°}$
(b) $50\ \angle 15°$
(c) $35.35\ \angle 15°$
(d) $70.7\ \angle 15°$

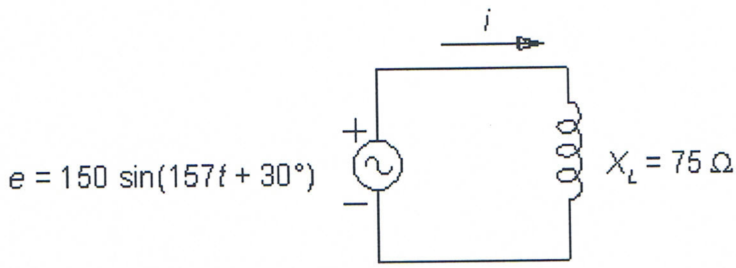
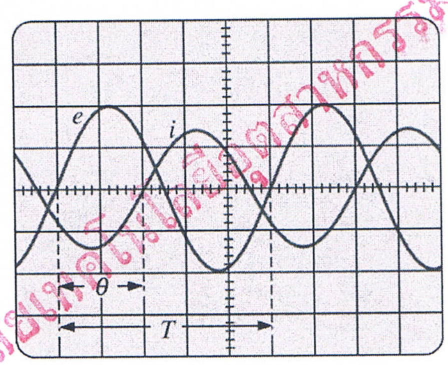


FIGURE 4

17. See Figure 4. What is the current i ?

(a) 1.5 sin(157t − 90°)
(b) 0.9 sin(157t + 60°)
(c) 2 sin(157t − 60°)
(d) 1.5 sin(157t + 60°)



Vertical sensitivity = 2 V/div.
Horizontal sensitivity = 0.2 ms/div.

FIGURE 5

18. See Figure 5. What relationship exists between voltages e and i?

(a) e lags i by 36°
(b) e leads i by 36°
(c) e lags i by 144°
(d) e leads i by 114°

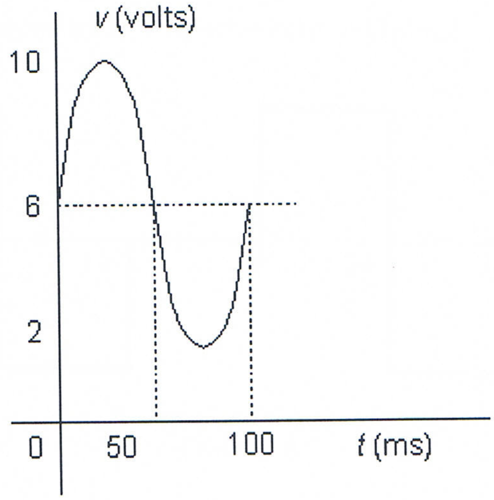


FIGURE 6

19. See Figure 6. What is the average value of the waveform?

    (a) 6 V (b) 2 V (c) 0 V (d) 5 V

20. See Figure 6. What is the frequency of the waveform?

    (a) 4 Hz (b) 10 Hz (c) 1 Hz (d) 2 Hz

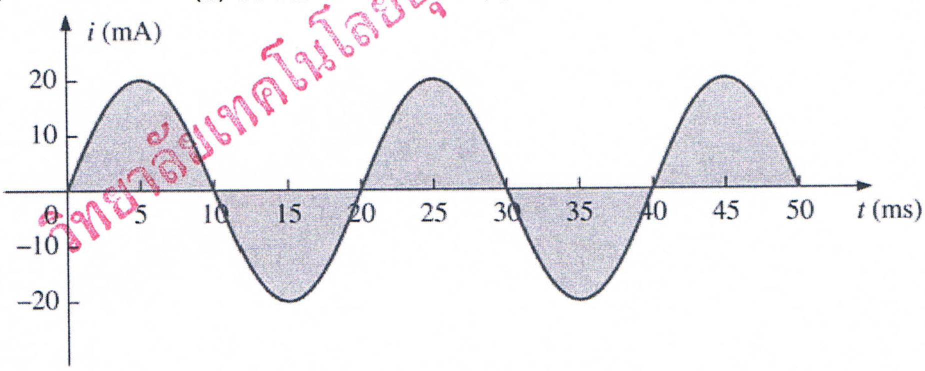


FIGURE 7

21. See Figure 7. What is the RMS value of the waveform?

    (a) 10 V (b) 12.72 V (c) 14.14 V (d) 28.28 V

22. See Figure 7. What is the average value of the waveform?

    (a) 0 V (b) 0.636 V (c) 6.36 V (d) 12.72 V

23. What is the phase relationship between the waveform v = 2 sin(140t + 27°) and i = 19 sin(140t − 43°)?

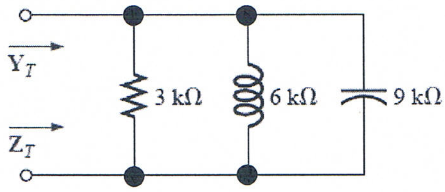
    (a) v leads i by 70° (b) i leads v by 55°.

    (c) i leads v by 70° (d) v leads i by 55°

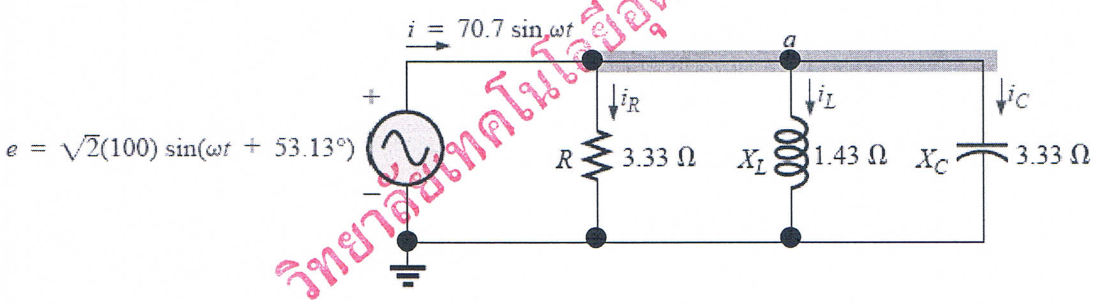
Part 3 (20 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

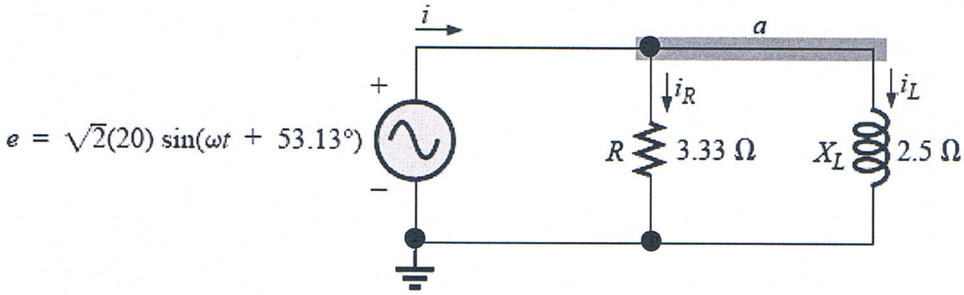
1. Find the total ADMITTANCE (Y_T) and IMPEDANCE (Z_T) of the circuit of below.



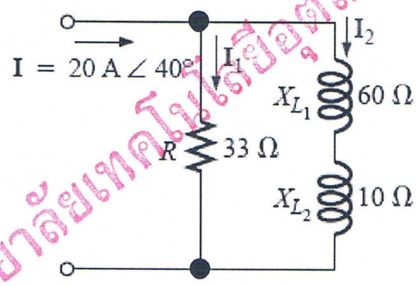
2. Draw the phasor diagram of the currents Is, I_R, I_L, and I_C, and the voltage E.



3. Find the average power (P) and power factor $\{\cos(\theta)\}$ of the circuit.



4. Calculate the currents I_1 and I_2 in phasor form using the current divider rule.



Asst. Prof. Dr. Poolsak Koseeyaporn
Asst. Prof. Dr. Titipong Lertwiriyaprapa

Answer Sheet

Name:____ID:____________________

Subject: 341152 Electric Circuits II

Part 1

	True	False
1.		
2.		
3.		
4.		
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Part 2

	a	b	c	d
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2.				
3.				
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26.				

24. Find the positive peak value of the waveform in Figure 8.

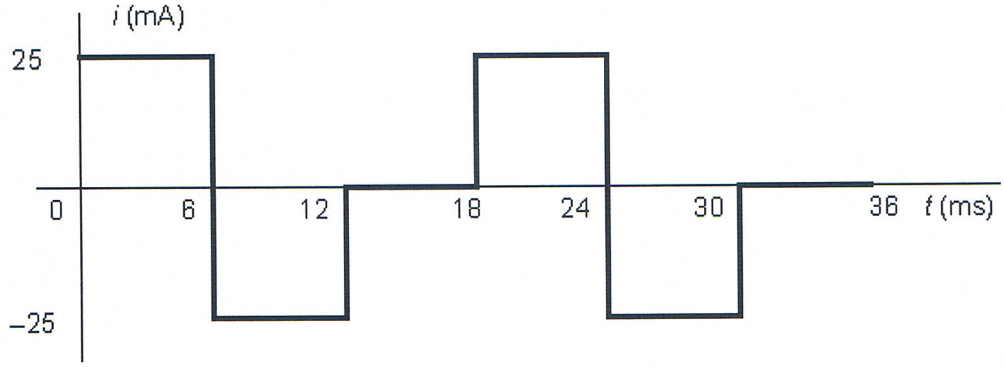

FIGURE 8

(a) 50 mA (b) 12.5 mA (c) 25 mA (d) 20 mA

25. Using the waveform in Figure 8, find the frequency of the current.

(a) 83.3 Hz (b) 167 Hz (c) 55.6 Hz (d) 0.083 Hz

26. Evaluate the expression $(4 + j5) \times (39 \angle -42°)$ and express the answer in polar form.

(a) $79.5 \angle 9.3°$ (b) $79.5 \angle -9.3°$ (c) $249.6 \angle 9.3°$ (d) $249.6 \angle -9.3°$